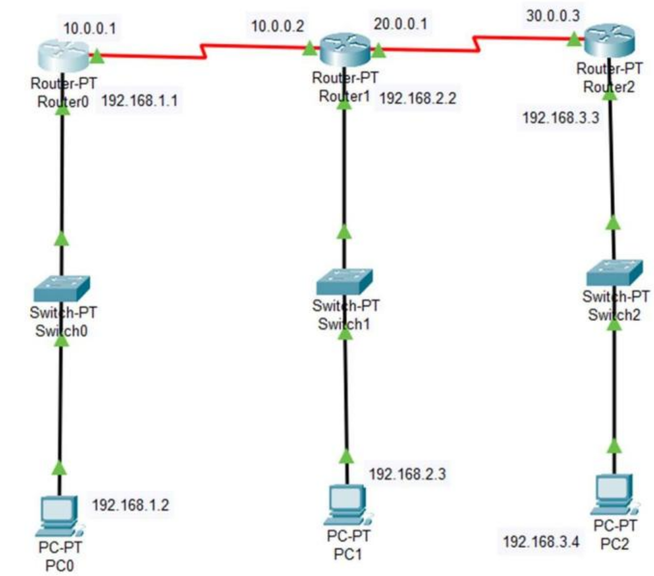
EXPERIMENT NO:-06

AIM:- To study about the configuration of distance vector RIP routing protocol by using Cisco Packet Tracer.

◉ **STEPS :**

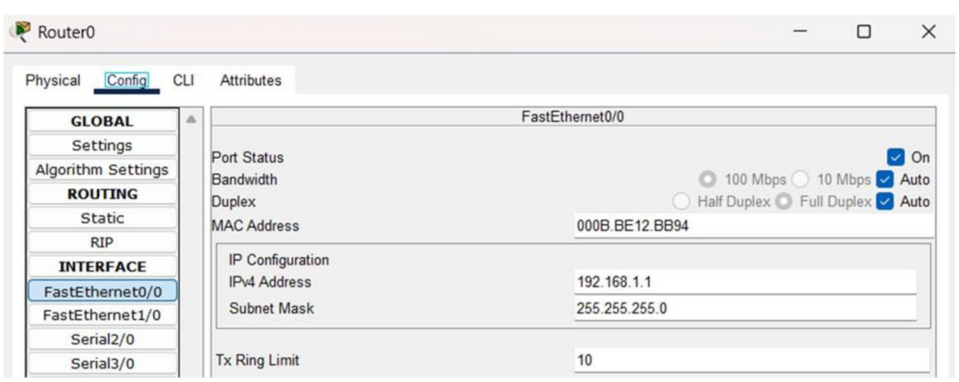
1. Setup the Network:

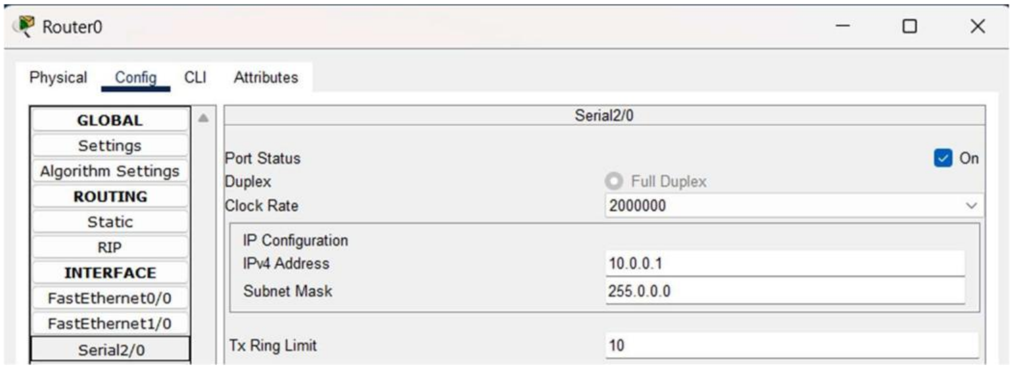
- Create a connection similar to the one shown in the reference diagram.
- Arrange routers, switches, and PCs, and connect them using appropriate cables.



2. Assign IP Addresses:

- Assign IP addresses to all routers and PCs step by step.
- For each PC, go to the **Desktop Tab > IP Configuration** and configure the IPv4 Address, Subnet Mask, and Default Gateway.





3. RIP Routing Configuration:

- Configure RIP on all routers.
- Use the following commands for each router:
  - On **Router0**:

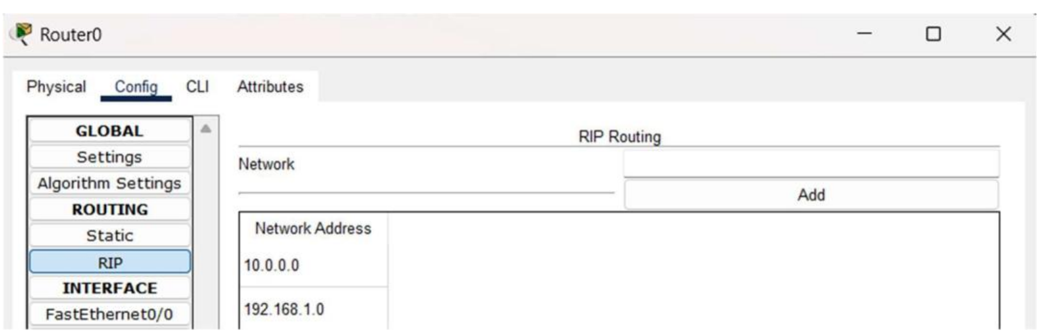
    ```
    router rip
    version 2
    network 192.168.1.0
    network 10.0.0.0
    ```

  - On **Router1**:

    ```
    router rip
    version 2
    network 10.0.0.0
    network 20.0.0.0
    network 192.168.2.0
    ```

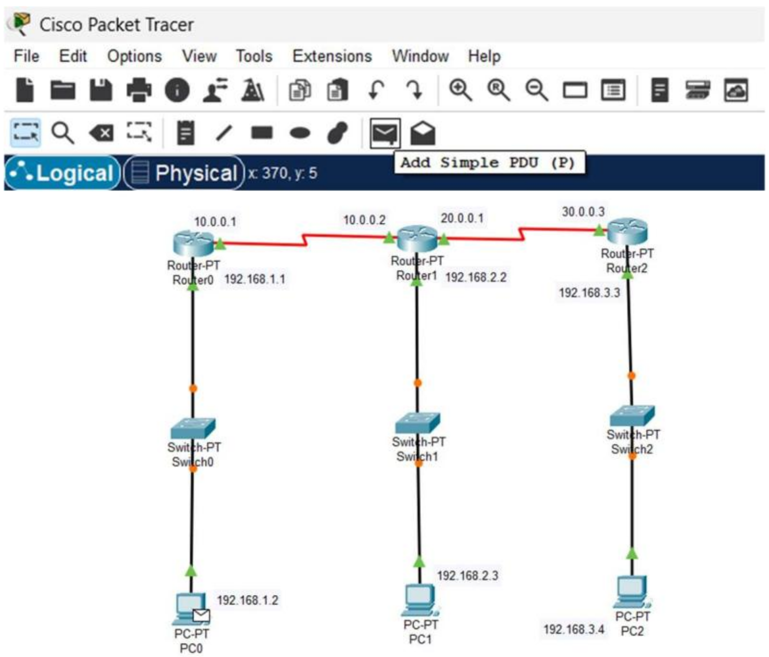
  - On **Router2**:

    ```
    router rip
    version 2
    network 20.0.0.0
    network 192.168.3.0
    ```



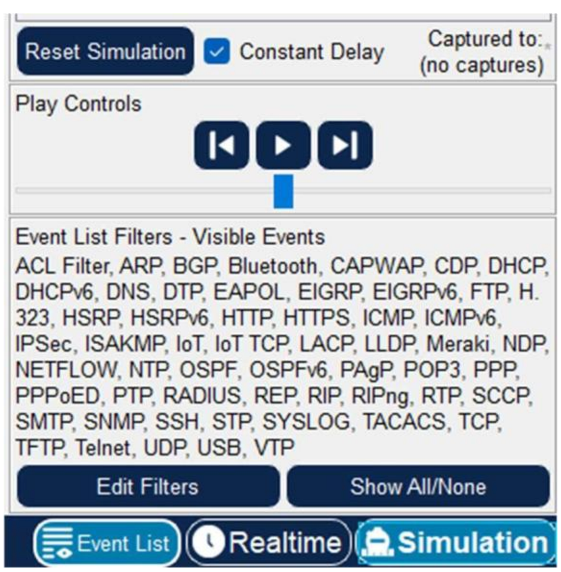
4. Send Data:

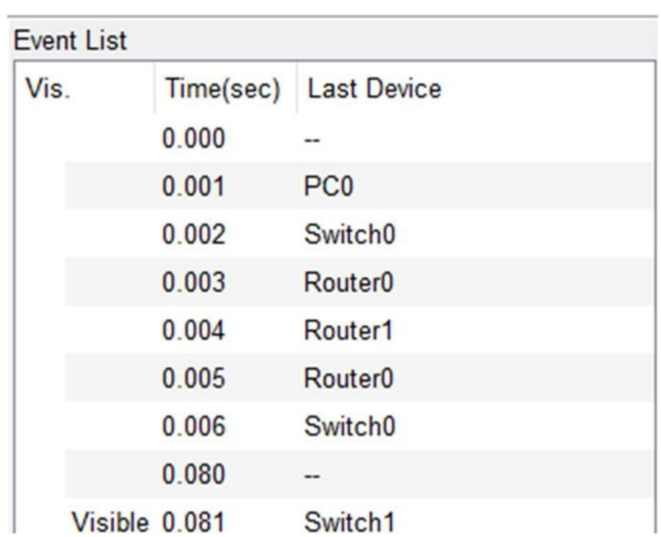
- Use the **Add Simple PDU** tool.
- Select a PC and send a message to a PC in a network connected to a different router (e.g., from PC0 to PC2).



5. Simulation Mode:

- Switch to **Simulation Mode**.
- Run the simulation to observe the RIP routing process and ensure successful packet delivery.
- Event List

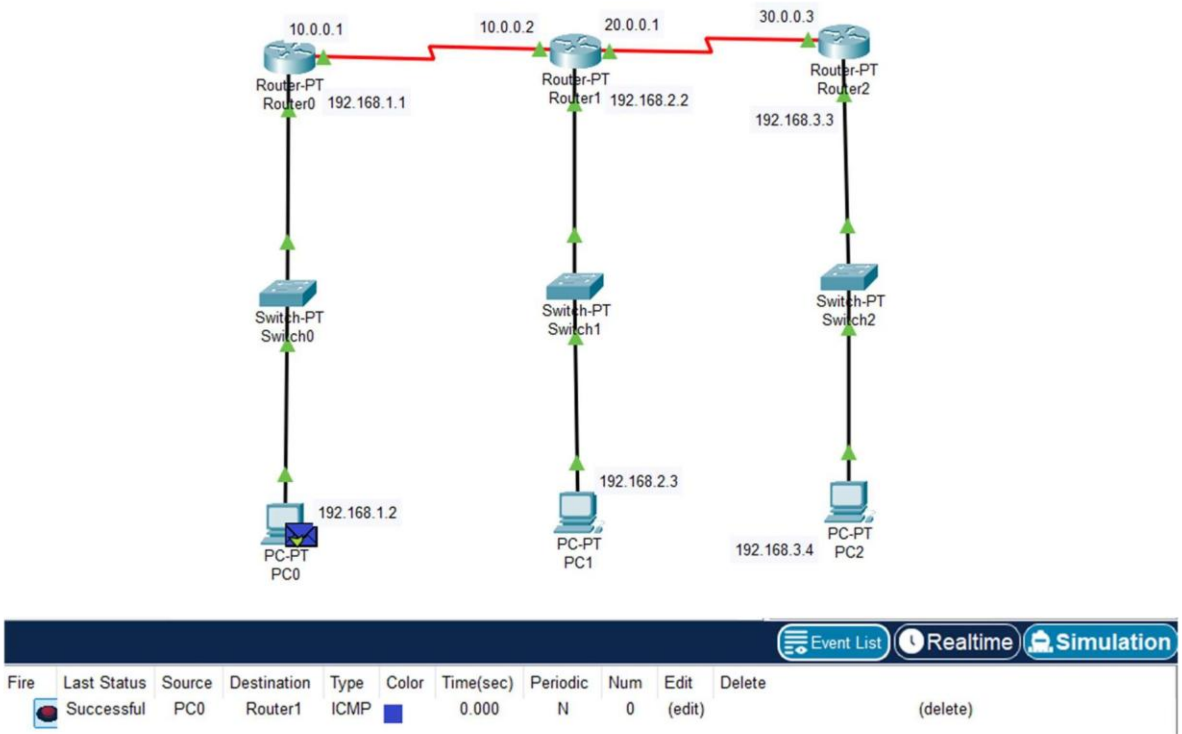




Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	Router0
	0.004	Router1
	0.005	Router0
	0.006	Switch0
	0.080	--
Visible	0.081	Switch1

6. After it done successfully we see like in shown the picture final step.



7. Verification:

- Open the **Command Prompt** on any PC and use the `ping` command to test communication between networks.
- Use commands like `ping` to verify the communication between devices on different networks.